

	Title	Data transmission protocol
	File	BM54_transmissionprotocol_rev01_20190805

Content

1.	Bluetooth Low energy.....	2
1.1	General Information.....	2
1.2	Advertising Data	2
1.3	Services and Characteristics.....	2
1.4	Blood Pressure Measurement.....	3
1.5	Blood Pressure Feature	4

1. Bluetooth Low energy

1.1 General Information

The BM54 is able to transfer stored measurement data via Bluetooth Low energy. The BM54 always works as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet. The newer version of BM54 uses 6 Digit PassKey Entry for the connection.

1.2 Advertising Data

The advertising name is "BM54" and the advertising data of the newer version contains the following information:

Name	Assigned Number	Description	Note
Manufacturer Specific Data	0xFF	0x110601 The first two bytes 0x0611 are the company ID of "Beurer". The 0x01 is used to identify devices that are using the passkey feature	

1.3 Services and Characteristics

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Generic Attribute	0x1801	00001801-0000-1000-8000-00805f9b34fb	
Blood Pressure	0x1810	00001810-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb	

Generic Access

Name	Assigned Number	Properties	Note
Device Name	0x2A00	Read	"BM54"
Appearance	0x2A01	Read	"0" - Unknown
Peripheral Privacy Flag	0x2A02	Read, Write	"0x00"
Reconnection Address	0x2A03	Write	
Peripheral Preferred Connection Parameters	0x2A04	Read	Connection Interval: 100,00ms – 200ms Slave Latency: 0 Supervision Timeout Multiplier: 1000

Generic Attribute

Name	Assigned Number	Properties	Note
Service Changed	0x2A05	Indicate	

Blood Pressure

Name	UUID	Properties	Note
Blood Pressure Measurement	0x2A35	Indicate	
Intermediate Cuff Pressure	0x2A36	Notify	Not used by BM54
Blood Pressure Feature	0x2A49	Read	Irregular Pulse Detection Feature supported

Device Information

Name	Assigned Number	Properties	Note
Manufacturer Name String	0x2A29	Read	Manufacturer Name
Model Number String	0x2A24	Read	Model Number
Serial Number String	0x2A25	Read	Serial Number
Hardware Revision String	0x2A27	Read	Hardware Revision
Firmware Revision String	0x2A26	Read	Firmware Revision
Software Revision String	0x2A28	Read	Software Revision
System ID	0x2A23	Read	0x 00 00 00 00 00 00 00 00

	Title	Data transmission protocol
	File	BM54_transmissionprotocol_rev01_20190805

IEEE 11073-20601 Regulatory Certification Data List	0x2A2A	Read	0x FE 00 65 78 70 65 72 69 6D 65 6E 74 61 6C
PnP ID	0x2A50	Read	Bluetooth SIG Company: Texas Instruments Inc. <0x000D> Product Id:0 Product Version: 272

1.4 Blood Pressure Measurement

The BM54 is sending the data automatically after a connection has been established successfully. This can be triggered by clicking the M1 or the M2 Button on the device. It is also triggered after every measurement. The device is sending all data from both users.

Blood Pressure Measurement – Indicate

For every measurement stored on the BM54 an indication is send.

Name	Format	Info	Note
flags	1 byte	See separate Flag table "flag description". Value is always 0x1E.	
systolic	sfloat (2byte)	The value of 1 equals 1,0 mmHg. The valid range is 0-300.	
diastolic	sfloat (2byte)	The value of 1 equals 1,0 mmHg. The valid range is 0-300.	
mean arterial pressure	sfloat (2byte)	Not supported by the BM54. The value is always 0.	
timestamp	time (7byte)	See table Timestamp description – Part of Blood Pressure Measurement for the timestamp data	
pulse rate	sfloat (2byte)	The value of 1 equals a pulse of 1. The valid range is 0-255.	
user ID	uint8 (1byte)	The value of 1 represents measurement from user 1 and the value of 2 measurements of user 2. In Bluetooth SIG documentation it starts with the value of 0.	
measurement status	2 byte		

Example:

Receiving: [0x1E 0x70 0x00 0x4D 0x00 0x00 0x00 0xDF 0x07 0x01 0x0E 0x0A 0x37 0x00 0x48 0x00 0x01 0x00 0x00]

Meaning: 0x1E – The flags set for the BM54 | 0x00 0x70 – Systolic value of 117 | 0x00 0x4D – Diastolic value of 77 | 0x00 0x00 – MAP value, not supported on the BM54 and always 0 | 0x07 0xDF – Year 2015 | 0x01 – Month January | 0x0E – Day 14 | 0x0A – Hour 10 | 0x37 – Minute 55 | 0x00 Second 0 | 0x48 00 – Pulse of 72 | 0x01 – User 1 | 0x00 0x00 - No Measurement error detected.

Flag description – Part of Blood Pressure Measurement

The flags have a size of 1 byte and are part of the blood pressure measurement Indication. The value for the BM54 is always 0x1E

unit flag	1 bit	systolic, diastolic and MAP unit: 0 = mmHg 1 = kPa	
time stamp flag	1 bit	0 = not present 1 = present	
pulse rate flag	1 bit	0 = not present 1 = present	
user ID flag	1 bit	0 = not present 1 = present	
measurement status flag	1 bit	0 = not present 1 = present	
reserved for future use	3 bit		

Timestamp description – Part of Blood Pressure Measurement

The timestamp has a size of 7 bytes and is part of the blood pressure measurement Indication.

Name	Format	Info	Note
year	uint16 (2 byte)	The value of 1 equals 1 year. The valid range is from 1582-9999	
month	uint8 (1 byte)	The valid range is from 0-12. The value 0 represents unknown, the value of 1 represents January, the value of 2 represents February ...	

	Title	Data transmission protocol
	File	BM54_transmissionprotocol_rev01_20190805

day	uint8 (1 byte)	The valid range is from 0-31. The value 0 represents unknown, the value of 1 represents the first day, the value of 2 represents the second day...	
hours	uint8 (1 byte)	The value of 1 equals 1 hour. The valid range is from 0-23	
minutes	uint8 (1 byte)	The value of 1 equals 1 minute. The valid range is from 0-59	
seconds	uint8 (1 byte)	The value of 1 equals 1 second. The valid range is from 0-59	

Measurement Status – Part of Blood Pressure Measurement

The measurement status has a size of 2 bytes and is part of the blood pressure measurement Indication.

Name	Format	Info	Note
body movement	1 bit	0 = no body movement 1 = body movement	
cuff fit	1 bit	0 = cuff fits properly 1 = cuff is too loose	
irregular pulse	1 bit	0 = no irregular pulse 1 = irregular pulse	
pulse rate range	2 bit	00 = in range 01 = exceeds upper limit 10 = below lower limit 11 = reserved for future use	
measurement position	1 bit	0 = proper position 1 = improper position	
reserved for future use	10 bit		

1.5 Blood Pressure Feature

Blood Pressure Feature – Read

The blood pressure feature tells the user which features are supported. The value for the BM54 is always [0x04 00]. That means the BM54 supports the irregular pulse.

Name	Format	Info	Note
body movement feature	1 bit	0 = feature not supported 1 = feature supported	
cuff fit feature	1 bit	0 = feature not supported 1 = feature supported	
irregular pulse feature	1 bit	0 = feature not supported 1 = feature supported	
pulse rate range	1 bit	0 = feature not supported 1 = feature supported	
measurement position	1 bit	0 = feature not supported 1 = feature supported	
multiple bond feature	1 bit	0 = feature not supported 1 = feature supported	
reserved for future use	10 bit		

	Title	Data transmission protocol
	File	BM54_transmissionprotocol_rev01_20190805

HISTORY	Changes to the previous document will be marked in red and documented below. Before updating the document, all previous red markings have to be set back to black colour.		Issued/ changed by
Revision	Date	Description	
01	2019-08-05	Initial version	MIMA